

Effect of heterogeneous risk perception on information diffusion, behavior change, and disease transmission

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Motivated by the importance of individual differences in risk perception and behavior change in people's responses to infectious disease outbreaks (particularly the ongoing COVID-19 pandemic), we propose a heterogeneous disease-behavior-information transmission model, in which people's risk of getting infected is influenced by information diffusion, behavior change, and disease transmission. We use both a mean-field approximation and Monte Carlo simulations to analyze the dynamics of the model. Information diffusion influences behavior change by allowing people to be aware of the disease and adopt self-protection and subsequently affects disease transmission by changing the actual infection rate. Results show that (a) awareness plays a central role in epidemic prevention, (b) a reasonable fraction of overreacting nodes are needed in epidemic prevention (c) the basic reproduction number R_0 has different effects on epidemic outbreak for cases with and without asymptomatic infection, and (d) social influence on behavior change can remarkably decrease the epidemic outbreak size. This research indicates that the media and opinion leaders should not understate the transmissibility and severity of diseases to ensure that people become aware of the disease and adopt self-protection to protect themselves and the whole population.

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I. INTRODUCTION

People's responses to infectious diseases could greatly affect the transmission patterns of diseases, and information about the transmissibility and severity of the disease conveyed through the media and opinion leaders plays a central role in raising awareness and influencing people's decision-making on whether or not to adopt self-protection, i.e., taking recommended practices to reduce the risk of infection, such as wearing a mask and washing hands with sanitizer in the COVID-19 context [1–7]. Here opinion leaders refer to individuals whose opinions are widely accepted by other people [8,9].

Many studies [10–16] used mathematical models to investigate how disease awareness affects the outbreaks of diseases. Most approaches [17,18] explored this problem by modifying the parameters in standard epidemic models. Funk *et al.* [19] first incorporated the effect of awareness into classic epidemic models and found that the spread of awareness could substantially reduce the epidemic outbreak size. Gross *et al.* [20] studied epidemic dynamics on an adaptive network in which susceptible nodes avoid contact with infected nodes by rewiring their connections. Wu *et al.* [21] modeled the effect of three forms of awareness: global awareness, local

awareness, and contact awareness. They showed that only global awareness cannot decrease the likelihood of an epidemic outbreak. Chen [22] examined how the amount of information affects behavior change and showed that increasing the amount of information that people possess may lower the likelihood of disease eradication.

Multiplex (also named multilayer) networks [23,24] have been developed to model the dynamic interactions between the spread of information and infection. The spreading of information and infection is represented by multiple network layers. For example, Granell *et al.* [25] proposed the susceptible-infected-susceptible unaware-aware-unaware model that can capture the critical point of the disease outbreak determined by the topological structure of the virtual information diffusion network. It can be extended to many model variants, such as the multiple-information model [26] that incorporates more than one type of information; the local awareness controlled contagion spreading model [27], in which the awareness transition is further influenced by the extent of the awareness of all neighbors; and the susceptible-infected-recovered (SIR) unaware-aware model [28] that also considers the recovered state. In addition to information diffusion and disease transmission, the transmission of protective behavior has been simulated [29].

Recent studies examined the effect of the media on infectious disease epidemics [30–37]. Liu *et al.* [32] proposed a mechanism to illustrate the effect of the media by

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